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BIOLOGY

1. Lacteals of villi absorb:

a) Amino acids

b) Glucose

c) Hydrochloric acid

d) Fatty acids

2. Microtubules perform all of the following except:

a) Transport of molecule

b) Movement in cilia and flagella

c) Movement of chromosomes

d) Protein synthesis

3. Which of the following is correct:

a) Glucose + fructose = maltose

b) Glucose + galactose = sucrose

c) Glucose + Glucose = maltose

d) Glucose + glucose = sucrose

4. Pepsin and trypsin:

a) Are both secreted with acid digestive juices

b) Are both secreted by the pancreas

c) breakdown protein to amino acids

d) are specific enzymes releasing peptides and polypeptide chains

5. Vitamin K:

a) is mostly found in green vegetables

b) Deficiency causes Pallegra

c) Promotes production of blood clotting factors

d) prevent night blindness

6. Regarding the respiratory system in humans:

A. The trachea is completely surrounded by rings of cartilage

B. The trachea divides into 3 bronchi

C. The vocal chords lie in the larynx

D. The lungs are rigid structures with little elasticity

7. The function of oxygen in cellular respiration is:

A. To combine with hydrogen from organic molecules to form carbon dioxide

B. To combine with carbon from organic molecules to form carbon dioxide

C. To oxidize ADP to ATP

D. To combine with glucose to form carbon dioxide

8. Which of the following increase the rate of respiration most efficiently?

A. Lack of oxygen in the alveoli

B. Excess of carbon dioxide in the blood

C. Excess of carbon dioxide in the lungs

D. Lack of oxygen in the tissues

9. The heart:

A. Pumps blood at a constant rate

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B. Oxygenate blood

C. Is under some amount of voluntary control

D. Is stimulated by SA node

10. All the following are true about the skin except:

A. It is the largest organ in the body

B. Vitamin E is synthesized here

C. It performs both secretary and excretory functions

D. Melanin is synthesized here

11. The motor nerve cell transmits impulses from:

A. The effector organ to the spinal cord

B. Receptor cells to the spinal cord

C. Receptor cells to the effector organ

D. Spinal cord to the effector organ

12. Removal of the thyroid gland in an adult would cause an increase in:

A. Basal metabolic rate

B. Conversion of glycogen to glucose

C. Excretion of Na^+2 from kidney

D. Secretion of TSH

13. Insulin causes:

A. Increased blood sugar

B. Increased calcium depositiok in bones

C. Increased permeability of cells to glucose

D. Increased cholesterol in blood

14. Myofibrils:

A. Are found in smooth muscles

B. Are the smallest fibres that make up a muscle

C. Are crossed by transverse Y-bands

D. Cannot contract

15. The mammalian placenta has all of the following functions except:

A. Excretion

B. Nutrition

C. Hormone Secretion

D. Stock absorptions

16. The following is a part of females reproductive tract:

A. Vas deferens

B. Vulva

C. Spermatozoa

D. Ovary

17. Where is oxygen produced in plants:

A. A root hair cell

B. Mesophyll cell

C. Lead epidermal cell

D. Phloem cell

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18. Rate of transpiration varies with all of the following except:

A. Temperature

B. Decreased number of chloroplasts

C. Light

D. Wind

19. The following are involved with plant respiration:

A. Stomata

B. Chloroplasts

C. Auxins

D. Geotropism

20. Nematoda Include all of the following except:

A. Hookworms

B. Guinea worms

C. Earthworms

D. Ascaries

21. Arthropods have all of the following except:

Hard Exoskeleton

A. Jointed Legs

B. Segmentation

C. Closed circulation

22. Rod shaped bacteria are called:

A. Bacilli

B. Plasmodia

C. Cocci

D. Spirilli

23. Most highly developed plants are:

A. Saprophytes

B. Angiosperms

C. Bryophytes

D. Gymnosperms

24. When hybrids are crossed the genotype of the offspring is:

A. 1:1

B. 3:1

C. 1:2:1

D. 4:1

25. The following may cause mutation:

A. Malaria

B. Tuberculosis

C. Ultraviolet Rays

D. Colchicines

26. Which one of the following is major advantage of using a light microscope instead if an electron microscope?

A. A superior resolving power

B. Constant depth of focus

C. Observation of living matter

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D. Use of very thin sections

27. Which one if the following posseses the most prominent golgi apparatus?

A. An islet of Langerhans

B. A retinal rod cell

C. A striated muscle cell

D. A xylem tracheid

28. Which of the following is a characteristic of mitochondria?

A. They store proteolytic enzymes

B. They provides sites for anaerobic respiration

C. They produce cellular Secretions

D. They release energy via Kreb's cycle

29. Polysaccharides are synthesized in plants by the process of:

A. Hydrolysis

B. Oxidation

C. Condensation

D. Glycolysis

30. Which one of the following is not immediately essential for protein synthesis:

A. ATP

B. Enzymes

C. Glucose

D. mRNA

CHEMISTRY

31. Which one of the following gas diffuses most rapidly?

A. Bromine

B. Carbon

C. Methane

D. Nitrogen

32. Moist ammonia gas may be dried by passing it through:

E. Calcium oxide

F. Calcium chloride

G. Calcium hydroxide

H. Conc: Sulphuric acid

33. Ordinary air from the atmosphere contains a out 1% oxygen, where as the production of oxygen in the mixture released by boiling river is 30%. The best explanation of the increase in percentage of oxygen is that?

A. Oxygen is more soluble in water than nitrogen

B. Carbon dioxide is more soluble in water than oxygen

C. Nitrogen reacts with water

D. The gases from boiled water contain no water vapour

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34. For two substances to be separated by peper chromatography it is necessary that:

A. They are both liquids

B. They are both soluble in same solvent

C. They have different densities

D. They have different colours

35. Which one of the following is not essential for photosynthesis:

A. Carbon dioxide

B. Chlorophyll

C. Sugar

D. Light

36. Crude Oil Can be Separated by Fractional distillation because it's constituents:

A. Are compounds

B. Are Chemically similar

C. Have different boiling points

D. Have different densities

37. Which one of the following is a pure compound?

A. Dry air

B. Ethanol

C. Steel

D. Tap water

38. Whose e/m ratio resembles with that of electrons?

A. Alpha rays

B. Beta Rays

C. Gamma rays

D. X-rays

39. What particles are present in solid lead(II) Bromide and molten lead(III) Bromide?

	Solid Lead Bromide	Molten Lead Bromide
A.	Molecules	Atom
B.	Molecules	Ions
C.	Ions	Atoms
D.	Ions	Ions

40. The ions X+ contains 23 particle in the nucleus and 10 electrons outside the nucleus. What does the nucleus of the ion X+ contain?

	Protons	Neutrons
A)	9	14
B)	10	13
C)	11	12
B)	12	11

41. The empirical formula of a liquid compound to be C2H4O. What other information is needed to work out it's molecular formula?

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A. The percentage composition of the compound

B. The relative molecular mass of the compound

C. The density of the compound

D. The volume occupied by 1 mole of the compound

42. Which statement concerning energy change is false?

A. Electrical energy is produced by the reaction in a battery

B. Light energy is absorbed during photosynthesis

C. Light energy is produced when methane is burned in oxygen

D. Heat energy is absorbed when Sodium reacts with water

43. A sample of air was bubbled into water. The pH of water slowly changes from 7 to 6. Which gas is sample caused this change?

A. Argon

B. Carbon dioxide

C. Carbon monoxide

D. Oxygen

44. Which statement about carbon monoxide is correct?

A. It is involved in photosynthesis

B. It produced carbon when burnt

C. It is a pollutant

D. It is denser than air

45. Which statement provides the best information to confirm that two substances are allotropic forms of carbon?

A. They had different crystalline structure

B. They both reduce heated Lead(II) Oxide to lead

C. Equal masses of both substances require equal masses of carbon dioxide and no other product when completely burnt in oxygen

D. Equal masses of both substances give equal masses of carbon dioxide and no other product when completely burnt in oxygen

46. What reacts at room temperature with ethanol and also with ethanoic acid?

A. Calcium carbonate

B. Copper(II) Oxide

C. Methanol

D. Sodium

47. Which statement concerning the ion of Group VII elements is correct?

A. Each contains an odd number of electrons

B. Each contains more protons than neutrons

C. Each contain more electrons than protons

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D. Each has 7 electrons in the outer most shell

48. Which formula represents the oxide of an element Y in Group II of the periodic table?

A. YO

B. YO2

C. YO3

D. Y2O

49. If four moles of sulphur dioxide are oxidized to sulphur Trioxide. How many moles of oxygen are required?

A. 0.5

B. 1

C. 2

D. 4

50. What is the number if paired shared electrons in methane molecule?

A. 1

B. 2

C. 3

D. 4

51. The relative molecular mass of carbon dioxide is 44. What is the mass of 24dm3 of carbon dioxide at STP?

A. 44

B. 22

C. 11

D. 12

52. When 0.24g of magnesium is reacted with excess HCl, what volume of gas is liberated at STP?

A. 120cm3

B. 240cm3

C. 360cm3

D. 3600cm3

53. which of the following gases is least common in air?

A. Argon

B. Carbon Dioxide

C. Nitrogen

D. Hydrogen

54. Which of the following elements is most abundant element in the universe?

A. Oxygen

B. Silicon

C. Hydrogen

D. Methane

55. Which of the following gases by the fractional distillation of liquid air?

A. Oxygen

B. Carbon dioxide

C. Hydrogen

D. Methane

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56. Which of the following is not a property of gamma particles?

A. They carry a charge of -1

B. The can penetrate aluminium foil about 15cm-20cm

C. They are a type of electromagnetic radiation

D. They have a mass approximately equal to a helium nucleus.

57. 10g of metal (relative atomic mass 40) are liberated by the passage of 0.5 moles of electrons through the molten chloride. What is the formula of chloride?

A. M2Cl

B. MCl

C. MCl4

D. MCl2

58. In which of the following instances is there no change in the concentration of the solution during electrolysis?

A. Concentrated HCl between carbon electrodes

B. Copper(II) sulphate solution between platinum electrodes

C. Copper(II) sulphate solution between copper electrodes

D. Dilute Sulphuric acid between Platinum electrodes

59. Which of the following correctly describes the process occuring at the electrodes when molten sodium chloride is electrolysed?

Anode	Cathode
A) Oxidation	Oxidation
B) Reduction	Reduction
C) Oxidation	Reduction
D) Reduction	Oxidation

60. Which one if the following compounds in a 1mol/dm3 solution has the lowest pH value?

E. Ammonia

F. Hydrochloric acid

G. Ethanoic acid

H. Sodium hydroxide

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PHYSICS

61. What are dimensions of work?

A. [MLT-1]

B. [MLT-2]

C. [ML2T-2]

D. [MLT]

62. Resultant of two forces 30N and 40N acting perpendicularly to each other is:

A. 70N

B. 10N

C. 50N

D. 40N

63. If the momentum of body is doubled. Its K.E will be:

A. Increased√2 times

B. Increased √3 times

C. doubled

D. quadrupled

64. First of all 'G' was measured in laboratory by:

A. Newton

B. Kepler

C. Cavendish

D. Plank

65. A body of mass 'm' is dropped from the top of a tower, during the first second it will fall through:

A. 9.8m

B. 4.9m

C. 98m

D. 49m

66. Angular displacement is measured in:

A. Radians

B. Meters

C. Foot

D. Pound

67. A simple pendulum is oscillating. When the bob is at its highest point, the string suddenly breaks, the bob will:

A. Keep in moving in circular path

B. Move horizontally

C. Move in a projectile motion

D. Move vertically downward with an acceleration of 9.8 ms^-2

68. A body is running along a circular path of circumference 44m and completes one round trip in 11 seconds. His displacement at the end of one trip is:

A. 4m

B. 44m

C. Zero

D. 484m

69. A ball 'A' is vertically dropped from the top of a building 49m high at the same time another ball 'B' is thrown horizontally with a speed of 20m/s. (neglecting the resistance of air)

A. Ball A will reach first than ball B

B. Ball Bi will reach first than ball A

C. Both balls will reach the ground at the same time

D. None of the above

70. If A x B=[[i,j,k],[2,3,4],[1,5,2]] then vector B is:

A. 2i + 3j + 4k

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B. i + 5j + 4k

C. -142 + 7k

D. 2i + 5j + 4k

71. Out of all existing four forces the weakest force is:

A. Gravitational force

B. Strong force

C. Electromagnetic force

D. Weak force

72. The relation between horse power and watt is:

A. 1hp = 547 watts

B. 1hp = 1000 watts

C. 1hp = 746 watts

D. 1hp = 476 watts

73. Velocity of sound in vacuum is:

A. 332m/s

B. 820m/s

C. 224m/s

D. Zero

74. One bel is equal to:

A. 10 decibel

B. 100 decibel

C. 1000 decibel

D. 0.1 decibel

75. In Newton's ring when interference pattern is viewed from the above by means of reflected light, the central spot always appears:

A. White

B. Black

C. Blue

D. Red

76. The frequency of second pendulum is:

A. 2Hz

B. 0.5Hz

C. 20Hz

D. 0.05Hz

77. The weight of a 20kg body at the center of earth is:

A. 19.6N

B. 196N

C. 0

D. 1.96N

78. If the distance between two bodies is doubled and their masses are also doubled, the gravitational force will:

A. Increase four times

B. Decrease four times

C. Remain constant

D. Increase two times

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79. The weight of an object on the moon is 30N. Its weight on the Earth will be:

A. 105N

B. 180N

C. 60N

D. 150N

80. The power of lens is measured in:

A. Watt

B. Joule

C. Diopter

D. Meter

81. SI unit of magnetic field is:

A. Gauss

B. Tesla

C. Weber/m-2

D. Ampere

82. The minimum charge on any object cannot be less than:

A. 1 coulomb

B. 1.6x10-19 C

C. 9.1x10-31 C

D. 8.8x10-12 C

83. The number of neutrons in 92U234 is

A. 92

B. 235

C. 143

D. 327

84. The most appropriate device used to measure the EMF of a cell is:

A. Voltmeter

B. Potentiometer

C. Avometer

D. Ammeter

85. Which quantity can be measured in the unit of Joule/coulomb?

A. Current

B. Potential difference

C. Power

D. Resistance

86. A 40 watt lamp turns half the electrical energy it uses into light energy. How much light energy does it give out in 10 seconds?

A. 400J

B. 200J

C. 800J

D. 20J

87. What is the smallest total resistance which can be obtained using only 6ohms and 12ohms resistors?

A. 2 ohms

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B. 4 ohms

C. 6 ohms

D. 8 ohms

88. The capacitor of 5uf and 10uf are connected in series. The total capacitance is:

A. 15uf

B. 5uf

C. 3.3uf

D. 1.11uf

89. Which of the following is correct answer?

A. 1F > 1C

B. 1F < 1C

C. 1F = 1C

D. None of the above

ENGLISH

90. The device which converts AC into DC voltage is called:

A. An amplifier

B. A rectifier

C. A Voltmeter

D. An oscillator

91. Because of the _____ and prolonged nature of the _____ water must be carefully conserved and rationed:

A. Arid, Reservoir

B. Dire, Forecast

C. Severe, Drought

D. Negligent, Emergency

E. Miserly, Supply

92. Generally, fund raising parties are quite labour intensive and not very cost effective in other words putting in a great deal of _____ doesn't mean you will _____ great deal of income:

A. Restraint, Obtain

B. Effort, generate

C. Successful, resistance

D. Challenged, discussion

E. Futile, Co-operation

93. WRITER : NOVELIST

A. Scientist : astronomer

B. Teacher : student

C. Physician : patient

D. Poet : searcher

E. Worker : Exertion

94. COW : CALF

A. Hog : pork

B. Horse : mule

C. Sheep : lamb

D. Tiger : Stripe

E. Ram : Eve

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95. If there is an error select the one underlined part that must be changed to make the sentence correct.

Everyone of the shops in the town **were closed(A)** on Tuesday **because(B)** of the **ten-inch(C)** rainfall **that had fallen**(D) during the day. **No Error (E)**

A

B

C

D

E

96. If there is an error select the one underlined part that must be changed to make the sentence correct.

Raising(A) living costs **together(B)** with escalating taxes, **have(C)** proved to be a burden **for everyone(D)**. **No Error(E)**

A. A

B. B

C. C

D. D

E. E

97. He is not Asking the questions. (Change the voice):

A. Not asking questions he is

B. Questions are not asked by him

C. Questions are not being asked by him

D. Not asking questions is what he is doing

98. Were the crimes confessed by the criminals? (Change the voice):

A. Was the confession by the criminals made?

B. Were the crimes committed confessed by the criminals?

C. Were the criminals confessing to the crimes?

D. Did the criminals Confess to the crimes?

99. Aneela said, "Shahana is not doing her work properly". (Change the voice into indirect speech):

A. Aneela say that Shahana did not do her work properly

B. Aneela said that Shahana was not doing her work properly

C. Shahana did not do her work properly was said by Aneela

D. She is not doing her work properly is what Aneela said about Shahana

100. Fahad said, "All that glitters is not gold". (Change the voice into indirect speech):

A. Fahad said that all that glitters is not gold

B. All that glitters is not gold said by Fahad

C. Fahad says that all that glitters is not gold

D. Fahad is saying for all that glitters is not gold

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ANSWER KEY

Q NO	ANS KEY	Q NO	ANS KEY	Q NO	ANS KEY	NO	ANS KEY
1	D	26	D	51	D	76	B
2	D	27	B	52	A	77	B
3	A	28	A	53	A	78	D
4	C	29	D	54	C	79	A
5	C	30	D	55	B	80	C
6	A	31	C	56	D	81	A
7	C	32	C	57	A	82	A
8	E	33	B	58	B	83	B
9	C	34	C	59	C	84	C
10	D	35	A	60	C	85	C
11	B	36	D	61	C	86	D
12	C	37	A	62	D	87	B
13	C	38	A	63	B	88	B
14	C	39	D	64	B	89	D
15	A	40	A	65	A	90	B
16	D	41	B	66	B	91	D
17	B	42	D	67	C	92	A
18	B	43	C	68	D	93	D
19	D	44	A	69	D	94	B
20	B	45	A	70	C	95	A
21	D	46	A	71	D	96	C
22	B	47	C	72	A	97	C
23	D	48	A	73	A	98	C
24	B	49	B	74	B	99	C
25	B	50	D	75	D	100	D

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